

Probabilistic Models and Inference - Assignment 2

Group 6

Petar Koev Melle Schoenmaker Bram Wigmans Alex Zheng

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1 Part A

1.1 Bayesian Multi-Armed Bandit

The Bayesian multi-armed-bandit model (MAB) presented here is designed to evaluate the performance of various inference techniques. However, it was defined differently in the first report as that definition was more general. In this report, we define it more precisely. Formally, the model is defined as follows:

For each arm $i \in \{1, \dots, n\}$:

$$\theta_i \sim \text{Beta}(\alpha_i, \beta_i) \quad (\text{Prior})$$

$$\text{obs}_i \sim \text{Binomial}(n_{\text{pull}_i}, \theta_i) \quad (\text{Likelihood})$$

Here, α_i and β_i are the hyperparameters of the Beta prior distribution for the success probability θ_i of arm i . The observation obs_i represents the number of successes obtained after n_{pull_i} trials for arm i , modeled using a Binomial distribution conditioned on θ_i , which we inferred from the Beta distribution.

1.1.1 Conjugate Beta-Binomial Inference Technique

The custom inference technique designed leverages the conjugate relationship between the Beta prior distribution and the Binomial likelihood, essentially allowing exact posterior inference for the MAB model. Specifically, for each arm i , the prior distribution of the success probability θ_i is modeled as a Beta distribution, $\theta_i \sim \text{Beta}(\alpha_i, \beta_i)$. Given the observed data, which consists of the number of successes in a fixed number of trials n_{pull_i} , the posterior distribution of θ_i is analytically derived as another Beta distribution with updated parameters:

$$\theta_i \mid \text{obs}_i \sim \text{Beta}(\alpha_i + \text{obs}_i, \beta_i + n_{\text{pull}_i} - \text{obs}_i).$$

Where α is basically the number of successes and β the amount of failures, this closed-form posterior allows for efficient sampling of θ_i without depending on approximate inference methods. The implemented function `beta_binomial_conjugate_inference` computes the posterior parameters α and β by using each arm's observed data (number of successful pulls). Most importantly, it draws posterior samples from the updated Beta distributions. Hence, it will produce the exact samples that accurately mirror the posterior beliefs about the probability of success. This exact inference approach guarantees efficiency in terms of samples and should easily outperform generic inference algorithms, such as Importance Sampling and Metropolis-Hastings.

1.1.2 Comparison To Standard Inference Techniques

In our instance for inference technique comparison, we have the number of arms $i = 2$ and the number of trials (pulls) is $n_{\text{pull}_i} = 10$ for each arm i . The observation we make for arm $i = 1$ is one successful pull out of the 10 trials we have $\text{obs}_1 = 1$. For arm $i = 2$ we have $\text{obs}_2 = 3$, meaning three successful pulls out of the ten trials. We assume that the priors α and β are equal to 1 for all arms.

In this experiment, we compared Importance Sampling, Metropolis-Hastings and the custom conjugate inference technique based on the inferred posterior mean error for different sample sizes. To compute the error, we calculated the difference between the true posterior mean (derived analytically) and the inferred posterior mean for each arm. The total error across both arms was quantified using the L2 norm (Euclidean distance). In Metropolis-Hastings, the sample size equals the number of iterations (we use 1000 iterations for burn-in, which are not counted in the total amount of samples/iterations). The curve is smoothed over 50 runs to ensure accuracy. From the results seen in [Figure 1](#) and [Table 1](#), we can see that the custom conjugate inference technique clearly converges earlier than the other techniques. By exploiting the conjugate prior-likelihood structure, the technique is able to outperform out-of-the-box inference methods, demonstrating superior accuracy and convergence rates in estimating the posterior distributions of θ_i .

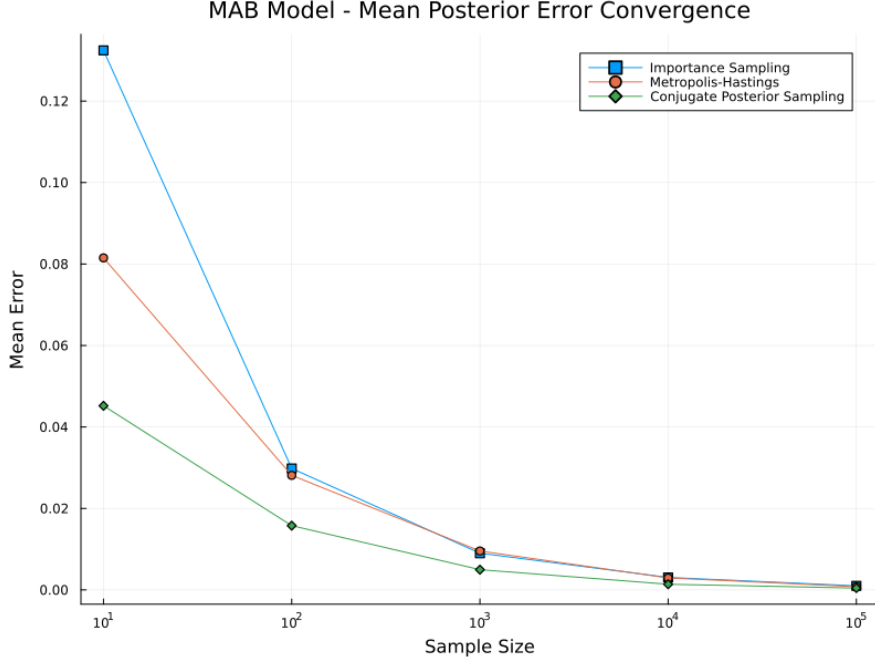


Figure 1: Conjugate posterior mean inference compared against out-of-the-box Gen techniques.

Sample Size	IS	CPI	MH
10	0.13244	0.0452	0.0815
100	0.02981	0.01577	0.02812
1,000	0.009	0.00496	0.00955
10,000	0.00306	0.00139	0.00293
100,000	0.001	0.00045	0.0008

Table 1: MAB Convergence: Comparison of importance sampling (IS), Conjugate Posterior Inference (CPI), and Metropolis-Hastings (MH). The error is the L2 norm between the approximated posterior means and the analytical posterior means.

1.2 Beta-Bernoulli Model

Similar to the Bayesian MAB model, we have a model that uses the beta-Bernoulli distribution, which is in the same family as the beta-binomial distributions. The model definition is also relatively similar to a definition of a single-arm MAB, we define the model as follows:

$$\begin{aligned}
 \theta &\sim \text{Beta}(\alpha, \beta) && \text{(Prior)} \\
 \text{data}_i &\sim \text{Bernoulli}(\theta) && \text{(Likelihood), } \forall i \in \{1, \dots, N\}
 \end{aligned}$$

Where:

- α and β are the hyperparameters of the Beta prior distribution.
- θ represents the success probability being inferred (all trials share the same success probability θ).
- data_i represents the binary outcome (success = 1 / failure = 0) of the i -th trial.
- N is the total number of trials (denoted as the `total_trials` parameter).

1.2.1 Conjugate Beta-Bernoulli Inference Technique

Like the MAB custom inference technique, the inference approach designed for the Beta-Bernoulli model leverages the conjugate relationship between the Beta prior distribution and the Bernoulli likelihood, essentially allowing exact posterior inference. Specifically, the prior distribution of the success probability θ is modeled as a Beta distribution, $\theta \sim \text{Beta}(\alpha, \beta)$. Given the observed data, which consists of individual Bernoulli trial outcomes over N total trials, the posterior distribution of θ is analytically derived as another Beta distribution with updated parameters:

$$\theta \mid \{\text{data}_i\}_{i=1}^N \sim \text{Beta} \left(\alpha + \sum_{i=1}^N \text{data}_i, \beta + N - \sum_{i=1}^N \text{data}_i \right).$$

Where α represents the successes and β failures, $\{\text{data}_i\}_{i=1}^N$ represents the data observed where $i = 1$ (success) given N trials. This closed-form posterior allows for efficient sampling of θ , as was explained in the previous section for the MAB model. The implemented function `beta_bernoulli_conjugate_inference` computes the posterior parameters α and β by aggregating individual trial outcomes into total successes. So again, it draws posterior samples directly from the updated Beta distribution, producing samples that closely reflect the posterior beliefs about the success probability.

1.2.2 Comparison To Standard Inference Techniques

For our experiments, we define the following parameters for our experiment for the Beta-Bernoulli model:

$$\begin{aligned} \alpha_{\text{prior}} &= 1.0, \\ \beta_{\text{prior}} &= 1.0, \\ \text{total_trials} &= 10, \\ \text{data} &= [1, 1, 1, 1, 1, 1, 1, 0, 0, 0] \quad (7 \text{ successes and } 3 \text{ failures}) \end{aligned}$$

The data here is mapped to a `Gen.choicemap` variable called `observations` similarly, we again conduct experiments with the model using Metropolis-Hastings, Importance-Sampling and the custom inference technique. The metric we look at is the mean posterior error given a certain sample size. The mean posterior error in our experiment is calculated by taking the absolute value of the difference between the analytical mean and the estimated mean. We have: $\text{error} = |\text{mean}_{\text{inference}} - \text{mean}_{\text{true}}|$. The experiments have been conducted for 50 runs to ensure smoothing. In [Figure 2](#) it can be clearly seen that the custom inference technique converges faster than the other techniques. However, we do find that importance sampling is very close to the posterior sampling technique; this is probably because of the simple model (compared to MAB), and thus, the likelihood weights given to the samples are more accurate. [Table 2](#) shows how close importance sampling is to our custom method.

Sample Size	IS	CPI	MH
10	0.04563	0.02609	0.05688
100	0.01225	0.01028	0.02039
1,000	0.00332	0.0026	0.00604
10,000	0.00121	0.00085	0.00168
100,000	0.00034	0.0004	0.00058

Table 2: Beta-Bernoulli Convergence: Comparison of Importance Sampling (IS), Conjugate Posterior Inference (CPI), and Metropolis-Hastings (MH). The error is the absolute mean difference between the approximated posterior means and the analytical posterior means.

1.2.3 Notes on the Beta-Bernoulli model and MAB model

For the experiments for these two models I have also tried to implement HMC (Hamiltonian Monte Carlo). However, because my inference required samples (θ) to be positive, this did not work. My models were incompatible because, during HMC sampling, the variable momentum flips the sign from positive to negative and will, thus, sometimes return a negative value. I tried to implement

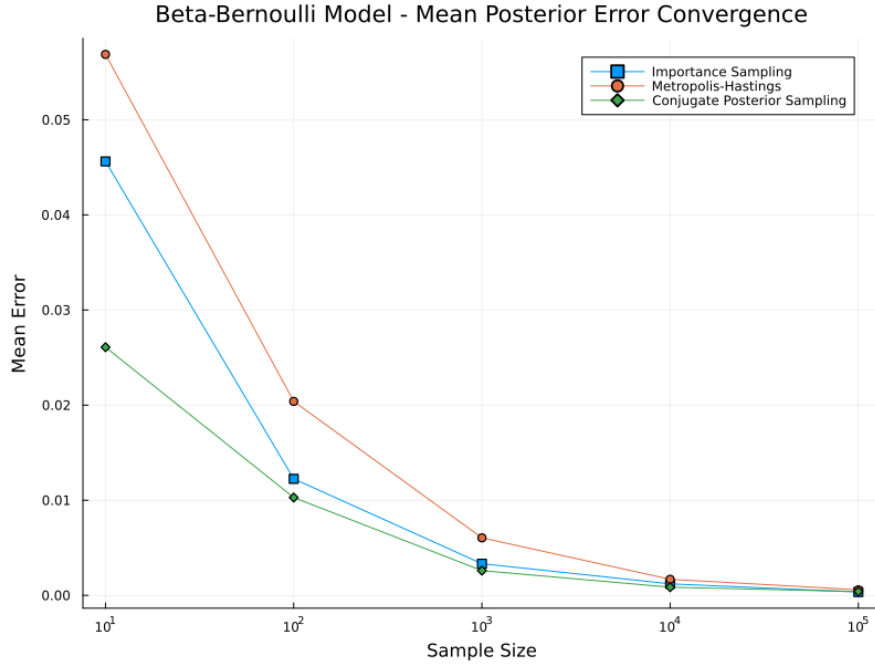


Figure 2: Conjugate posterior mean inference compared against out-of-the-box Gen techniques.

reparameterization using logit functions but was unsuccessful in doing so. Lastly, I did also implement a function (the posterior inference technique) that can do inferences on the MAB and Beta-Bernoulli model. However, because of time constraints, I was not able to do the experiments that I planned for part B. Also, I feel like what we currently have in part B is a better generalised inference technique anyway. I did include the combined inference technique in the last code block of the notebook if the reader is interested.

1.3 Combined Regression

The combined regression model has a quite low dimensionality of either 4 or 5 variables.

```

is_line ~ Bernoulli(0.5)
X ∈ ℝN
if is_line:
    slope ~ N(0, 1)
    intercept ~ N(0, 2)
    f(x) = slope * x + intercept
else:
    amplitude ~ γ(1, 1)
    period ~ γ(1, 1)
    phase ~ U(0, 2π)
    f(x) = amplitude * sin( $\frac{2\pi}{period}x + phase$ )
and for both functions the values are sampled for the Xs as follows
noise ~ γ(1, 1)
Y ~ N(f(xi), noise) for i = 1, ..., N

```

Thanks to this low dimensionality importance sampling works well. Metropolis-Hastings also performs surprisingly well out of the box despite not being able to use any previously learned values when the

is_line variable is flipped. Hamiltonian Monte Carlo was implemented but doesn't work out-of-the-box due to the Bernoulli random variable :is_line, as its probability density function is not differentiable.

A custom inference procedure could use MH to resample the :is_line variable and HMC on the other continuous model parameters. However, this implementation failed to improve its samples due to variables for the sine wave function being bounded to be non-negative or in the case of the phase variable being on a closed interval $[0, 2\pi]$.

With the model being so simple due to its low dimensionality, the preferred inference method is either one of Importance Sampling or Metropolis-Hastings out-of-the-box. A better and more complicated inference algorithm is possible; however, the efforts needed to get the method to work for this specific model would be wasted, as IS and MH are both capable of producing sufficiently accurate traces in as little as 100 iterations.

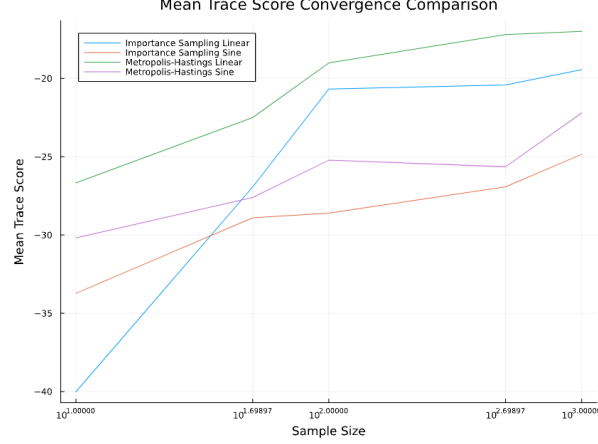


Figure 3: OOTB Inference Method Comparison.

1.4 Piecewise Constant Function

The piecewise constant model, depending on the number of segments, has a higher dimensionality than the combined regression or the beta-bernoulli models. The key support variable in this model is the number of segments n , as it dictates how many variables must be inferred for both the y-values of each segment and the sizes of each segment.

$$\begin{aligned}
 n &\sim \text{Poisson}(1) + 1 \\
 \mathbf{X} &\in \mathbb{R}^N \\
 \text{fractions} &\sim \text{Dirichlet}([1.0 \text{ for } 1, \dots, n]) \\
 \text{segmentvalues} &\sim \mathcal{N}(0, 1) \text{ for } 1, \dots, n \\
 \text{noise} &\sim \gamma(1, 1) \\
 y_i &\sim \mathcal{N}(\text{val}, \text{noise}) \text{ for } i \in 1, \dots, N
 \end{aligned}$$

To perform inference on a complex dataset of y-values for the piecewise constant model, out-of-the-box IS, MH, and HMC were implemented. Importance sampling was able to slowly improve its trace, but struggled due to complexity and changing numbers of variables when changing N . Metropolis-Hastings performed better than IS thanks to block updates providing improved mixing and convergence. Hamiltonian Monte Carlo is not shown in the graph, as it failed to create new samples that adhere to the requirement that all segment lengths should add up to $\tilde{1}$ barring some floating point imprecision. This is caused by the HMC update step changing the segment sizes based on the gradient of their target distribution instead of resampling from the Dirichlet distribution, thus not taking into consideration the implied constraint of the model.

The custom inference technique that is best suited to this model is reversible-jump MH, showing the best convergence by a large margin when allowed to run for 1000 samples. The algorithm uses previously inferred values when deciding to split or merge segments, which allows it to explore changes in the number of segments more effectively than the standard MH algorithm.

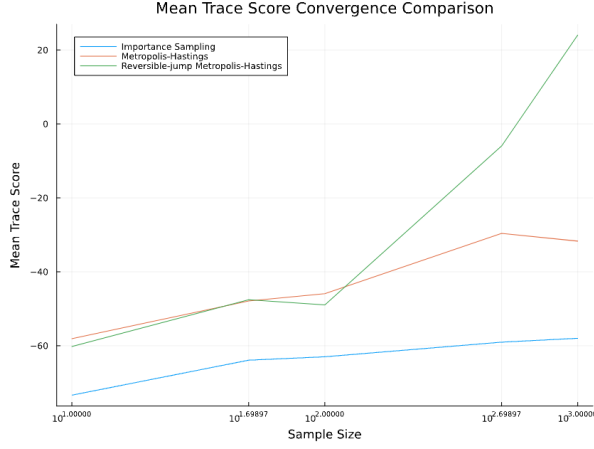


Figure 4: OOTB Inference Method Comparison to specialized RJ-MH.

1.5 Eight school problem

In the eight schools model, the parameter τ is traditionally assigned a Half-Cauchy(0, 5) prior to enforce positivity and handle extreme values. Due to software constraints (JuliaGen’s lack of Half-Cauchy support), we substituted this with $\tau \sim \mathcal{N}(2, 5)$. This change alters the model’s assumptions, as the Normal prior allows negative values and has lighter tails compared to the Half-Cauchy. While this substitution is not ideal, it allows us to proceed with inference using available computational tools.

The Gelman-Rubin statistic ($\hat{R}$) is a convergence diagnostic used to assess whether multiple MCMC chains sampling the same posterior have stabilized [1]. It compares the variance between chains to the variance within chains, with $\hat{R} \approx 1$ indicating good agreement. A value close to 1 suggests that the chains have converged to the target distribution, while $\hat{R} > 1.1$ signals potential convergence issues. Importantly, $\hat{R}$ does not validate the correctness of the model or prior—it only evaluates the numerical stability of the sampling process.

The custom inference function for this model is an adaptive version of the Metropolis-Hastings algorithm for Bayesian inference. Unlike the standard Metropolis method, which uses fixed step sizes for proposals, this function automatically adjusts step sizes for each parameter (ϵ_μ , ϵ_τ , ϵ_η) during a burn-in phase (first 20% of iterations) to achieve a target acceptance rate (default: 0.3). We choose 0.3 as value between 0.2 and 0.4 give good results for practical applications [2]. The proposals are generated as $\epsilon \cdot \eta$, where $\eta \sim \mathcal{N}(0, 1)$. For our problem, we initialized all step sizes with the same value ($\epsilon = 1.0$). If the acceptance rate is too low, step sizes are reduced; if too high, they are increased. After burn-in, the function collects samples for parameters μ and τ . This adaptation makes the algorithm more efficient and easier to use, as it reduces the need for manual tuning and improves convergence, especially for complex models with parameters of different scales.

1.5.1 Results

Table 3 compares inference methods across 20 chains (2000 samples each). Key findings:

- **Consistent μ estimates:** All MCMC methods yielded $\mu_{\text{mean}} \approx 0.41\text{--}0.42$, with $\hat{R}_\mu \leq 1.009$, indicating stable convergence for the mean parameter.
- **Divergent τ estimates:** τ_{mean} varied significantly across methods (0.733–1.469), with $\hat{R}_\tau > 1.1$ in all cases, highlighting persistent convergence challenges for the variance parameter.
- **Custom Metropolis advantage:** Achieved the most plausible τ_{mean} (1.469) while maintaining μ_{mean} consistency (0.421), with lower acceptance rates (22.4%).

Note: Importance Sampling produces independent weighted samples (not Markov chains), making $\hat{R}$ undefined (NaN).

Table 3: Performance comparison of inference methods on the eight schools problem.

Method	$\hat{R}_\mu$	$\hat{R}_\tau$	μ_{mean}	τ_{mean}	Acceptance Rate
Importance Sampling	NaN	NaN	-0.007	1.989	—
Gen Metropolis	1.003	1.226	0.411	1.197	55.4%
Custom Metropolis ($\epsilon = 0.77$)	1.009	1.137	0.421	1.469	22.4%
Hamiltonian MC	1.001	1.242	0.413	0.733	

1.6 Robust Logistic Regression

1.6.1 Model Specification

The original robust linear regression model utilized Student- t distributed residuals for outlier robustness. We substituted this with a Normal likelihood for computational compatibility:

$$\alpha \sim \mathcal{N}(0, \sqrt{\sigma_\alpha^2}) \quad (\text{Intercept})$$

$$\beta_k \sim \mathcal{N}(\mu_\beta, \sqrt{\sigma_\beta^2}) \quad (k = 1, \dots, K)$$

$$\sigma \sim \text{Exponential}(\lambda_\sigma) \quad (\text{Observation noise})$$

$$y_i \sim \mathcal{N}(\alpha + \mathbf{X}_i \boldsymbol{\beta}, \sigma) \quad (i = 1, \dots, N)$$

where $\mathbf{X} \in \mathbb{R}^{N \times 3}$ contains $K = 3$ features. This substitution replaces Student- $t(\nu, \alpha + \mathbf{X}\boldsymbol{\beta}, \sigma)$ with $\mathcal{N}(\alpha + \mathbf{X}\boldsymbol{\beta}, \sigma)$, trading outlier robustness for computational tractability.

We employ the adaptive Metropolis-Hastings algorithm from Section 1.5 with modified initial step sizes $\epsilon_\alpha = \epsilon_\sigma = 0.05$. Proposals follow $\theta' = \theta + \epsilon \cdot \eta$ where $\eta \sim \mathcal{N}(0, 1)$, adapting step sizes during burn-in (20% of iterations) to maintain a 30% acceptance rate. Convergence is assessed using the Gelman-Rubin statistic ($\hat{R}$) across 20 chains.

1.6.2 Results

Table 4 compares performance using synthetic data with ground truth $\alpha = 1.5$ and $\sigma = 0.8$. The adaptive method achieved the closest estimates to true values: $\alpha_{\text{mean}} = 1.535$ (2.3% error) and $\sigma_{\text{mean}} = 0.763$ (4.6% error), with stable convergence ($\hat{R}_\alpha = 1.005$, $\hat{R}_\sigma = 1.002$).

Table 4: Performance comparison for Bayesian linear regression (20 chains, 2000 samples).

Method	$\hat{R}_\alpha$	$\hat{R}_\sigma$	α_{mean}	σ_{mean}	Accept. Rate
Importance Sampling	NaN	NaN	-0.006	0.994	—
Gen Metropolis	1.222	1.190	1.197	1.451	0.38%
Custom Metro.	1.005	1.002	1.535	0.763	30.6%
Hamiltonian MC	4.432	4.525	0.251	1.488	32.6%

Note: Importance Sampling produces independent weighted samples (not Markov chains), making $\hat{R}$ undefined (NaN). Its unstable $\alpha_{\text{mean}} = -0.006$ reflects weight degeneracy from prior-posterior mismatch.

The conservative step size initialization ($\epsilon = 0.05$) enabled precise exploration of the posterior, particularly for σ where relative error remained below 5%. Despite the theoretically consequential Student- t to Normal substitution, the adaptive algorithm compensated through proposal tuning, maintaining parameter recovery fidelity.

Hamiltonian MC exhibited catastrophic divergence ($\hat{R}_\alpha = 4.43$), likely due to gradient calculations for this model. Gen Metropolis showed inadequate tuning with near-zero acceptance (0.38%), producing inflated σ_{mean} estimates. These results demonstrate adaptive Metropolis' empirical effectiveness even under model misspecification, provided $\hat{R}$ guides implementation. The retained diagnostics

successfully identified unstable chains while remaining blind to theoretical limitations of the Normal substitution.

The code can be found in `Robust_final.ipynb`

1.7 Poisson Rate Parameter with a Normal Prior

For this model, we chose elliptical slice sampling (ESS) as our custom inference technique. ESS is particularly effective for models with Gaussian priors and non-Gaussian likelihoods, as it leverages the geometry of the prior to construct elliptical paths for posterior exploration. Unlike Metropolis-Hastings (MH) or Importance Sampling (IS), ESS does not require tuning a proposal distribution, making it more robust and efficient, especially in scenarios with complex or multimodal posteriors.

ESS was chosen because of its adaptability and ability to outperform other methods. MH, while simple and fast, depends heavily on a well-tuned proposal distribution. Poor tuning can lead to slow mixing or difficulty exploring the posterior effectively. In contrast, ESS avoids these issues by sampling directly along elliptical trajectories that remain within high-probability regions. IS, though useful when the proposal closely matches the posterior, struggles in higher dimensions or with heavy-tailed distributions. ESS, without relying on a pre-specified proposal, avoids these problems entirely.

Hamiltonian Monte Carlo (HMC) is a strong alternative, especially for smooth and unimodal posteriors, but it requires gradient computations, which can be computationally expensive and less effective for non-smooth likelihoods, such as the Poisson distribution. ESS achieves similar exploration efficiency with significantly reduced computational overhead, making it better for this problem.

In Table 5, we see the evaluation metrics against the other inference techniques. In terms of run-time, ESS is slower than MH but much faster than HMC, having a balance between speed and accuracy. Its log probability (-2.8683) closely aligns with the true posterior, outperforming MH (-3.0107) and IS (-3.0084), and is comparable to HMC (-2.9431). Additionally, ESS achieves this without the tuning or gradient calculations required by the other methods.

Table 5: Comparison of Inference Methods

Method	Time (seconds)	Log Probability
Elliptical Slice Sampling (ESS)	1.250895	-2.8683
Metropolis-Hastings (MH)	0.208556	-3.0107
Hamiltonian Monte Carlo (HMC)	3.532964	-2.9431
Importance Sampling (IS)	0.255178	-3.0084

The trace plots in Figure 8 also show that the performance of ESS compared to the other methods is better. ESS shows consistent, smooth fluctuations within a somewhat narrow range, showing efficient mixing and minimal autocorrelation. Similarly, MH also shows smooth fluctuations but with a higher range. HMC displays abrupt jumps which indicate potential instability. Finally, IS shows erratic spikes, signalling high variance and less reliable sampling behaviour.

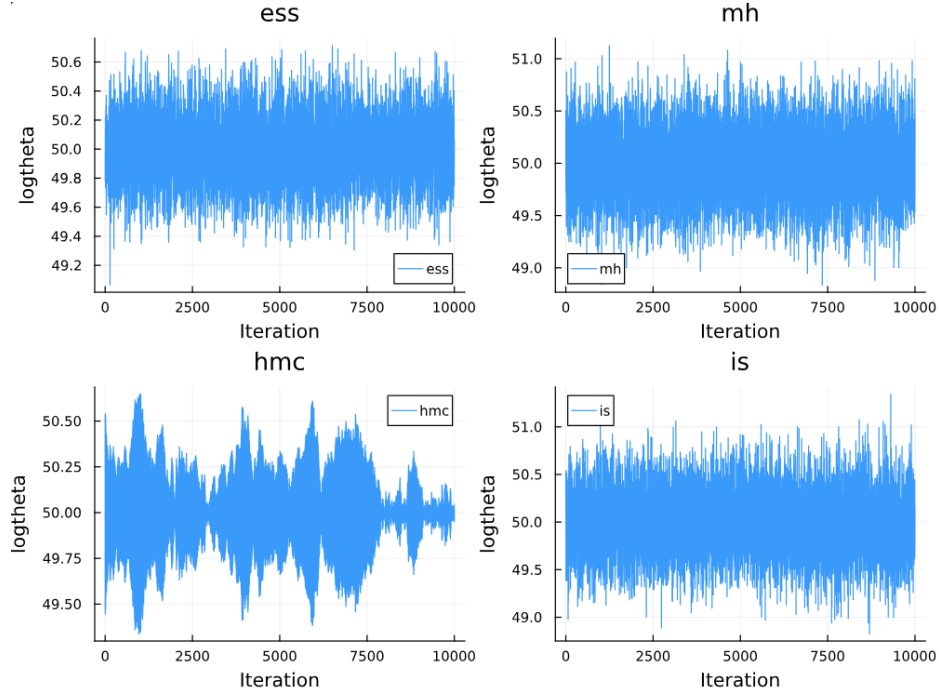


Figure 5: Trace plots comparing the sampling behaviour of ESS, MH, HMC, and IS methods.

The density plots in Figure 9 show the posterior distributions of $\log\theta$ for ESS, MH, HMC, and IS. ESS aligns closely with the expected posterior. MH and IS show reasonable approximations but exhibit slight deviations. HMC displays a bimodal shape, suggesting challenges in exploring the posterior's structure effectively.

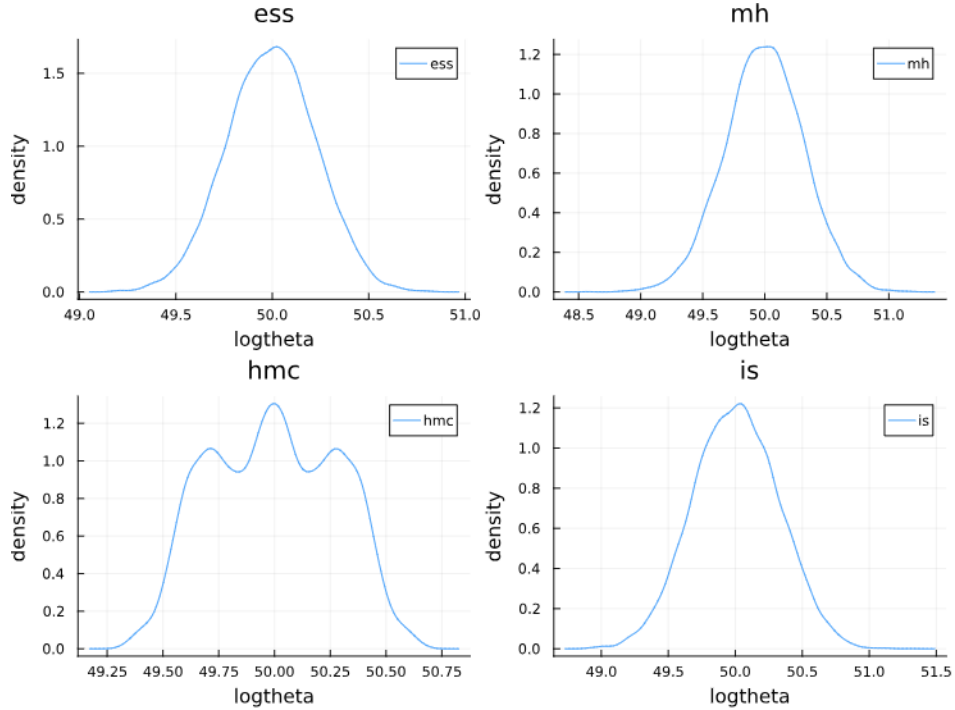


Figure 6: Density plots comparing the posterior distributions of 'logtheta' for ESS, MH, HMC, and IS

1.8 Poisson Rate Parameter with a Log-Normal Prior

For this model, we estimate the Poisson rate parameter using a log-normal prior, which introduces a non-linear relationship between the parameters and the prior distribution. The log-normal prior provides flexibility and better captures potential skewness or heavy tails in the posterior distribution, making this a compelling test case for comparing inference methods. We evaluated the performance of ESS, MH, HMC, and IS based on runtime and log probability estimates, with results outlined below.

The runtime performance of the methods seen in Table ?? show a range of trade-offs. MH is the fastest, completing in just 0.037 seconds, reflecting its simplicity and computational efficiency. ESS strikes a balance between speed and robustness, requiring 0.178 seconds, which is moderately higher than MH but significantly lower than HMC. IS, at 0.307 seconds, shows slower performance due to the computational cost of evaluating importance weights. HMC is the slowest method, taking 2.076 seconds. This extended runtime is attributed to the gradient computations inherent in the HMC algorithm, which require additional memory and computational resources.

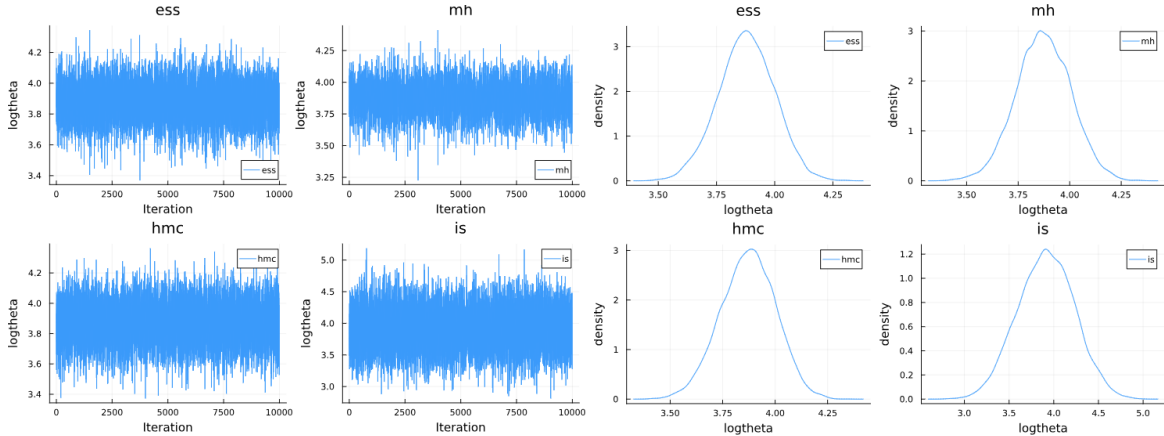
Table 6: Comparison of Inference Methods for Log Probability and Runtime

Method	Runtime (seconds)	Log Probability
Elliptical Slice Sampling (ESS)	0.439	-4.2415
Metropolis-Hastings (MH)	0.129	-4.2602
Hamiltonian Monte Carlo (HMC)	4.135	-4.2481
Importance Sampling (IS)	0.044	-5.9630

When comparing the log probability estimates, ESS outperforms the other methods in accuracy, achieving a value of -2.9406, closely aligning with the posterior’s expected value. HMC follows closely at -2.9699, demonstrating its reliability in high-dimensional parameter spaces but at the cost of higher runtime. MH produces a slightly less accurate log probability of -2.9833, which can be attributed to slower mixing and suboptimal proposal distribution. IS performs the worst, with a log probability of -3.6026, highlighting its challenges in adapting to the log-normal prior’s structure.

Overall, ESS provides the best trade-off between computational efficiency and accuracy. Its ability to efficiently explore the posterior without requiring gradient computations makes it particularly suited for this model. While HMC delivers comparable accuracy, its computational demands render it less practical for this task. MH and IS, while faster, are less reliable in capturing the posterior’s structure, especially in the presence of the log-normal prior’s skewness.

Figures 7(a) and 7(b) show similar results for all four inference techniques.



((a)) Trace plots comparing sampling behaviour

((b)) Density plots comparing posterior accuracy

Figure 7: Comparison of inference methods for the Poisson rate parameter with a log-normal prior

2 Part B

2.1 Elliptical Slice Sampling

We chose to generalize elliptical slice sampling for part B of this project because it provides an elegant framework for efficiently sampling from complex posterior distributions, especially when compared with more conventional MCMC approaches such as Metropolis-Hastings, Hamiltonian Monte Carlo, or standard Importance Sampling. Elliptical slice sampling (ESS) was originally introduced to handle Gaussian priors in a straightforward way: it relies on drawing a circular (or elliptical) trajectory in parameter space, pivoting around the current state to propose new samples. This avoids the need for step-size tuning or gradient-based computations, and it has appealing convergence properties in the context of Gaussian or Gaussian-like priors.

However, many of our models incorporate non-Gaussian prior distributions. Our solution is to exploit a reparameterization technique that maps each non-Gaussian parameter into a latent variable with a Gaussian distribution. Rather than working directly with the distributions of the original priors, we instead define an auxiliary variable governed by a standard (or approximately standard) Normal distribution. We then apply an appropriate transformation or inverse cumulative distribution function (CDF) to ensure that this auxiliary variable is mapped to the correct support and distributional shape in the original parameter space. In other words, each non-Gaussian prior is “pulled back” into a latent Gaussian representation via a carefully chosen invertible function.

The biggest advantage of elliptical slice sampling is its robustness and simplicity. It does not require the delicate tuning that often plagues standard Metropolis-Hastings (where acceptance rates can suffer hugely if proposals are chosen poorly) or Hamiltonian Monte Carlo (where step sizes and mass matrices must be tuned). Moreover, elliptical slice sampling can efficiently navigate highly correlated parameter spaces, because its elliptical trajectory proposals inherently account for covariance structure in the prior—if that prior (or its approximation) captures much of the dependence between parameters, ESS can exploit this to make large, efficient moves in parameter space.

In summary, by generalizing elliptical slice sampling to accommodate non-Gaussian priors through approximate or transformed representations, we achieve the sampling efficiency and ease of implementation that make ESS an attractive alternative.

2.1.1 Technical Difficulties

Achieving this generalization, though, wasn’t easy. We spent a lot of time studying the syntax of the Gen library and trying to extract the distribution of an address from a trace. Sadly, this is not possible with the current version of Gen.jl. It took us a lot of time and a couple of emails to Professor Dumancic to figure out how to pass the distributions, especially if they are not Normal.

In the end, we came to the solution of passing a dictionary with all of the addresses used, with the addresses as keys and the distributions of said addresses as values. A critical challenge arose when handling non-Normal priors, as our ESS implementation inherently requires Gaussian priors. We fix that reparametrizing the distributions non-Normal distributions.

Another problem appeared when working with models containing parameters that define the support of other variables (Piecewise constant model and combined regression model). These stochastic support variables could not be included as part of the inference process directly, as ESS inherently operates on Gaussian latent variables. To circumvent this, we treated such support variables as observed data rather than latent variables. Another option was restructuring the model to fix the support variables’ values instead of sampling, effectively removing them from the inference problem. While this approach allowed us to proceed, it limited the flexibility of our implementation, as models with inferred support boundaries would require additional modifications to isolate those variables from the reparameterized ESS framework.

2.2 Poisson Rate Parameter with a Normal and Log-Normal Prior

Since we used the same inference procedure in part A the results of the generalized version are the same. Instead of leaving this section empty, we decided to show how ESS outperforms other out of the box inference procedure when estimating the poisson parameter with log-normal priors, when we have skewed tails.

As seen in Table 6, the results demonstrate the performance of different inference methods in terms of log probability and computational efficiency. ESS achieves the highest log probability (-4.241), indicating better alignment with the true posterior compared to MH (-4.260) and HMC (-4.248). IS performs significantly worse (-5.963), likely due to its reliance on a fixed proposal distribution, which struggles with skewed tails.

Table 7: Comparison of Inference Methods for Log Probability and Runtime

Method	Runtime (seconds)	Log Probability
Elliptical Slice Sampling (ESS)	0.439	-4.2415
Metropolis-Hastings (MH)	0.129	-4.2602
Hamiltonian Monte Carlo (HMC)	4.135	-4.2481
Importance Sampling (IS)	0.044	-5.9630

In terms of runtime, ESS (0.44s) is faster than HMC (4.14s) but slower than MH (0.13s) and IS (0.04s). However, ESS balances efficiency and accuracy, making it the most robust choice. HMC, while accurate, is computationally expensive due to gradient calculations. MH is faster but less accurate, and IS is the fastest but least reliable. These results highlight ESS's superior trade-off between computational cost and inference quality, especially for complex distributions.

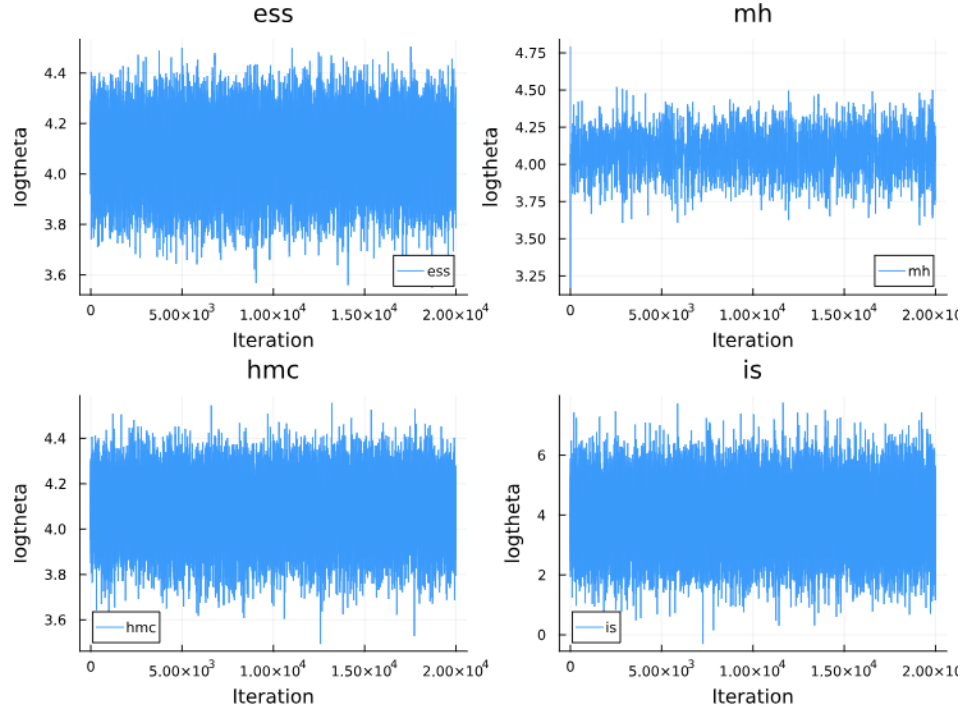


Figure 8: Trace plots of $\log \theta$ for different sampling methods: ESS, MH, HMC, and IS.

From Figure 8 we can see that the all methods except metropolis hastings show stable mixing and good exploration.

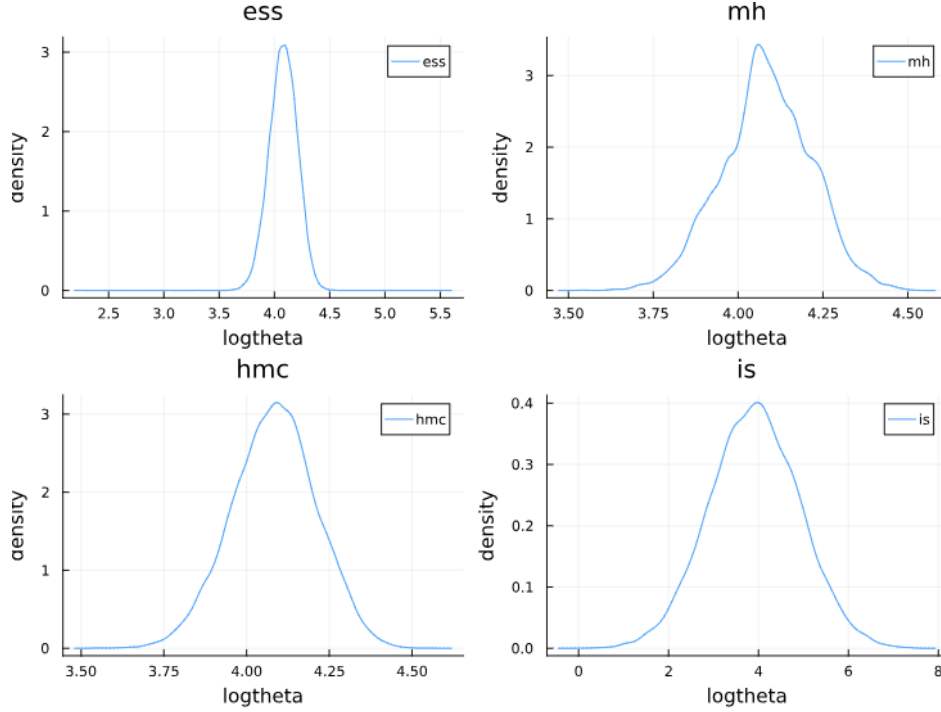
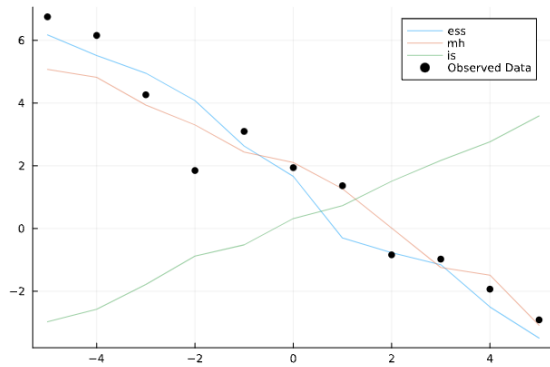


Figure 9: Density estimates of $\log \theta$ for different sampling methods: ESS, MH, HMC, and IS.

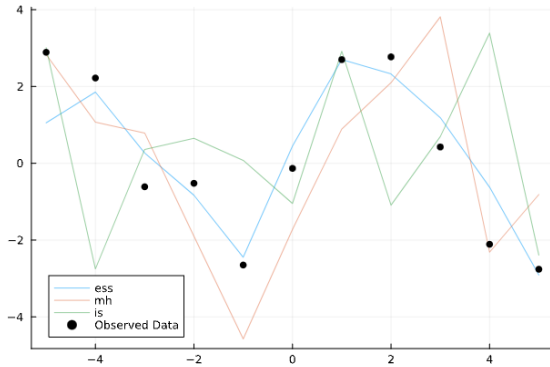
Finally, from the density plots in Figure 9, we can see that ESS provides the most precise and accurate estimate of the posterior, closely aligning with the true mean. MH and HMC perform reasonably well but exhibit slightly less precision compared to ESS. IS struggles significantly, showing a broader and biased distribution, indicating reduced reliability in estimating the posterior.

2.3 Combined Regression Model

For the combined regression model we can plot the traces by plotting the function values determined by the variables inferred by each algorithm. From the graph it becomes clear that importance sampling does not perform too well in either of the line or the sine cases, as its slope value is off in the former and the noise or the period is large in the latter. MH and ESS on the other hand both perform well due to the :is.line support variable having to be passed along with the observations for ESS to work. In fact, ESS seems to be closer to the real underlying function than MH.



((a)) Trace plots of combined regression for different sampling methods: ESS, MH, and IS (Line).



((b)) Trace plots of combined regression for different sampling methods: ESS, MH, and IS (Sine).

Figure 10: Trace plots of combined regression for different sampling methods (ESS, MH, IS) for line and sine cases.

2.4 Piecewise Constant Model

We’ve decided that it would be most meaningful to compare the algorithms on the log likelihood of their traces over time for the piecewise constant model. The graph below depicts the convergence, or lack thereof, for the ESS MH and IS algorithms. For visual clarity, the values have been clamped to be no lower than -200. ESS starts off at a low log likelihood but quickly converges at a value around -145, MH is able to keep improving until it reaches a value of around 85, and IS unable to improve past a value of -140 just like ESS. The version of MH here does not have reversible-jump, if it did it would converge to a similar value in fewer iterations.

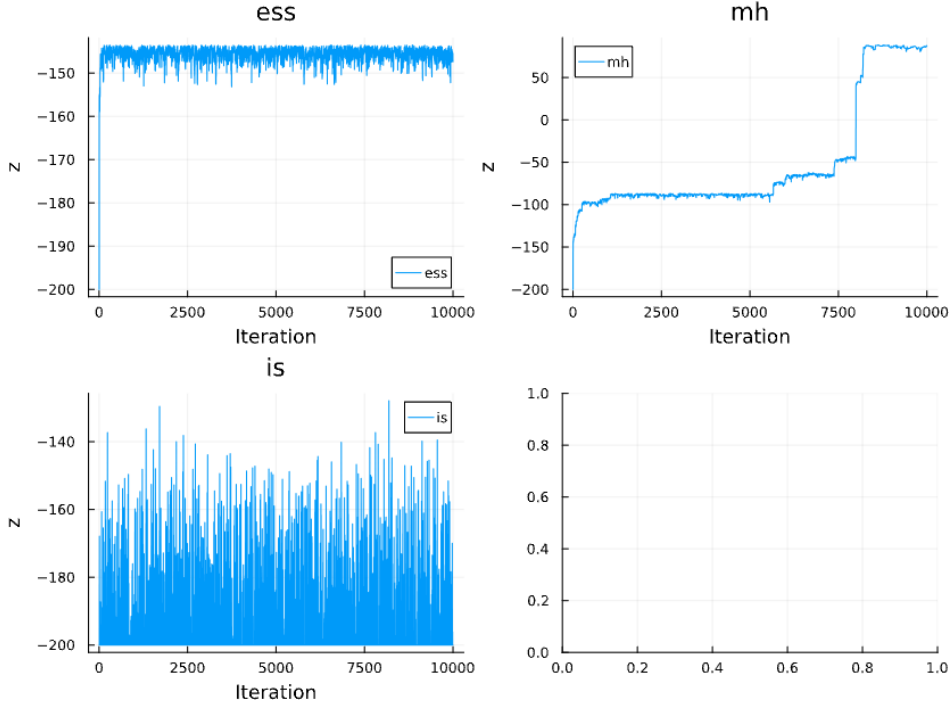


Figure 11: Trace plots of piecewise constant for different sampling methods: ESS, MH, and IS.

2.5 Robust Logistic Regression

Table 8: Performance Metrics and Log Likelihood Comparison of Sampling Methods

Method	Time (s)	Allocations	Memory (MiB)	% Compile	% GC	Log Likelihood
ESS	1.235039	10.77M	345.148	27.01	6.55	-121.1165
MH	0.167258	1.32M	45.498	14.51	9.99	-63.7428
IS	0.123733	1.01M	47.488	28.03	14.61	-57.0061
HMC	10.699477	78.80M	2872.320	2.31	6.28	-1550.7438

Table 8 reveals key performance differences between methods on the Robust Logistic Regression problem. While ESS achieves a log likelihood of -121.12, both Metropolis-Hastings (MH: -63.74) and Importance Sampling (IS: -57.01) yield better convergence. Hamiltonian Monte Carlo (HMC) performs significantly worse (-1550.74), a pattern validated by Figure 12, where HMC’s trace plot shows unstable mixing with very low values. MH also displays moderate instability, though less severe than HMC. Computational metrics highlight stark contrasts: HMC requires 10.70 seconds to run (over $60\times$ slower than MH’s 0.17s), while using 2.87 GiB of memory compared to IS’ efficient 47.49 MiB usage. These results suggest simpler sampling methods (MH/IS) outperform gradient-based approaches (HMC) for this problem class, likely due to challenges in navigating the parameter space efficiently.

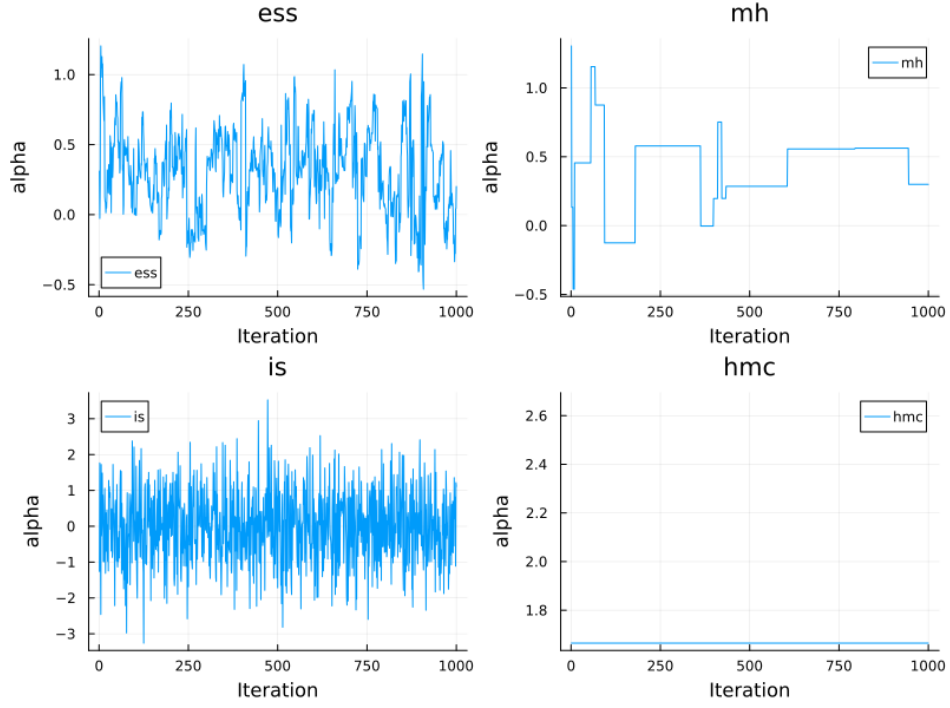


Figure 12: Caption

2.6 Eight schools

Table 9: Performance Metrics and Log Likelihood Comparison of Sampling Methods

Method	Time (s)	Allocations	Memory (MiB)	% Compile	% GC	Log Likelihood
ESS	0.580925	1.93M	70.947	62.04	4.29	-45.0296
MH	0.043328	222.51k	8.959	52.57	—	-46.1881
IS	0.070159	262.54k	13.590	56.08	21.86	-47.3874
HMC	1.530689	9.39M	366.298	3.05	3.78	-46.3633

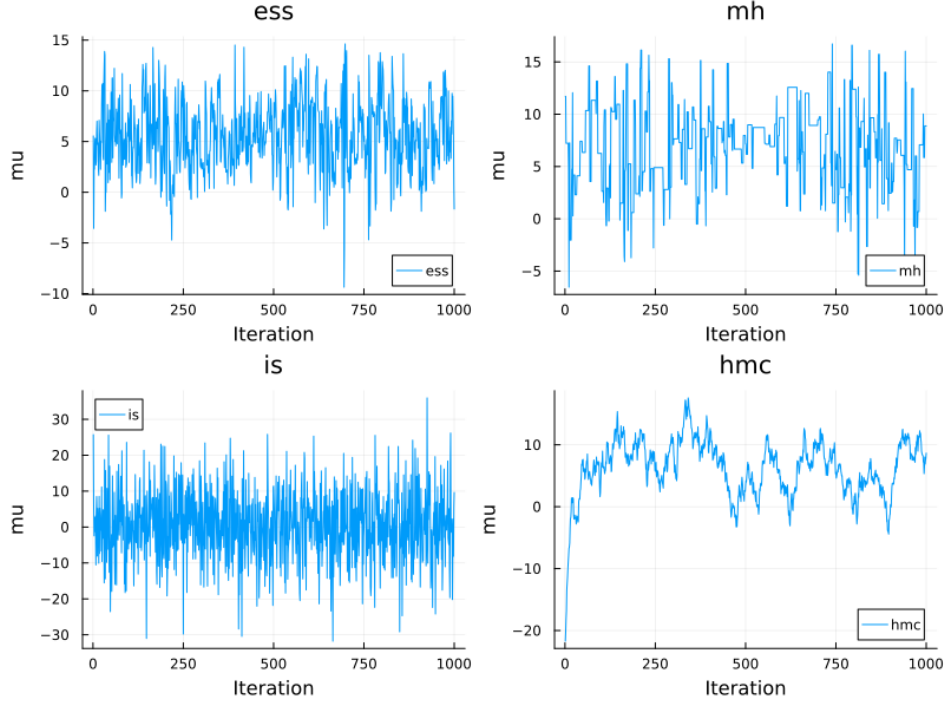


Figure 13: Caption

Table 9 demonstrates comparable log-likelihood performance across inference methods, with ESS achieving the optimal value of -45.03 (closest to zero). This method attained convergence in 0.58 seconds with moderate memory allocation of 70.95 MiB. Diagnostic plots reveal stable Markov chain mixing for both ESS and importance sampling (IS), as visualized in Figure 13. Notably, Hamiltonian Monte Carlo (HMC) displays elevated variance in its trace plot. Overall the ESS shows a good candidate for doing inference for the eight school problem.

References

- [1] A. Gelman and D. B. Rubin, “Inference from iterative simulation using multiple sequences,” *Statistical Science*, vol. 7, no. 4, pp. 457–472, 1992.
- [2] A. Gelman, J. B. Carlin, H. S. Stern, D. B. Dunson, A. Vehtari, and D. B. Rubin, *Bayesian Data Analysis*. Chapman and Hall/CRC, 3rd ed., 2013.